

Customer & Loan Portfolio Analysis

Assessing loan portfolio health and risk to support
smarter lending decisions

Prepared by: Eva Arliana
Data Analyst Portfolio Project I 2026



Business Context

Understanding the lending portfolio landscape.



Loan Portfolio

Understanding portfolio size, composition, and outstanding exposure.



Credit Risk

Monitoring repayment quality and identifying concentration of risk exposure.



Lending Decisions

Using portfolio insights to support risk-based underwriting and monitoring.



Understanding portfolio health and risk exposure is essential for making informed lending decisions.

Business Problem

The Challenge: Turning Loan Data Into Actionable Insights.



How can RevoFin identify portfolio risk patterns and understand the loan characteristics associated with weaker repayment performance?

Portfolio Health



Assessing portfolio performance across cohorts and key risk metrics.

Risk Concentration



Identifying segments contributing disproportionately to delinquent exposure.

Risk Drivers



Understanding how loan characteristics relate to repayment performance.

Business Questions & Objectives

Turning the business challenge into analytical questions.

Business Questions

- 1 Which loan cohort shows weaker repayment performance?
- 2 Where is delinquent exposure concentrated across risk segments?
- 3 How do loan characteristics such as grade, interest rate, and loan amount relate to repayment performance?
- 4 What customer-level insights can be reliably derived from the available data?
- 5 What actions can help reduce risk exposure and strengthen lending decisions?

Objective

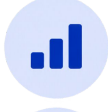


Assess portfolio health, identify key risk concentrations and loan characteristics associated with weaker repayment performance, and translate the findings into actionable lending recommendations.

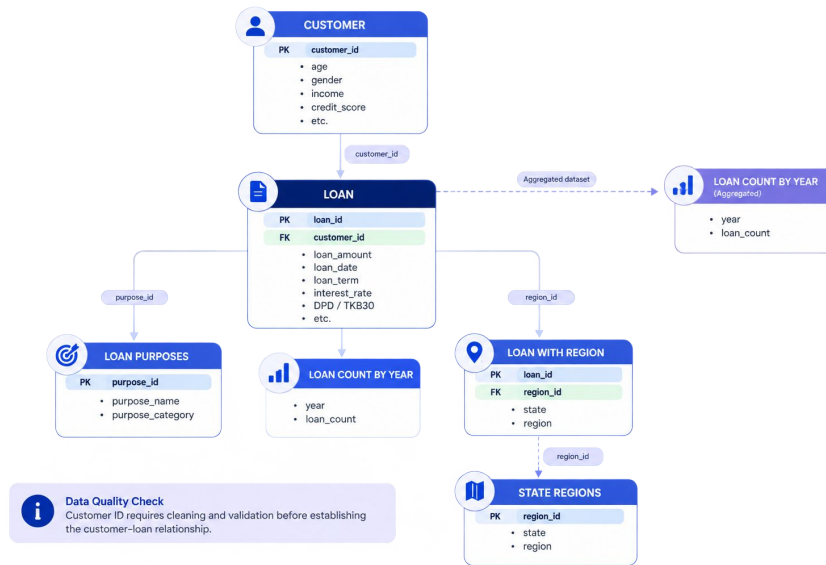
Datasets & Data Model

Six datasets provide complementary views of customers, loans, purposes, and regions.

Available Datasets (6)

-  customer.csv → Customer profile → Limited analytical use
-  loan.csv → Core loan & risk analysis → Primary
-  loan_count_by_year.csv → Temporal trend → Supporting
-  loan_purposes.csv → Loan purpose analysis → Exploratory
-  loan_with_region.csv → Regional analysis → Exploratory
-  state_regions.csv → Region mapping/reference → Reference

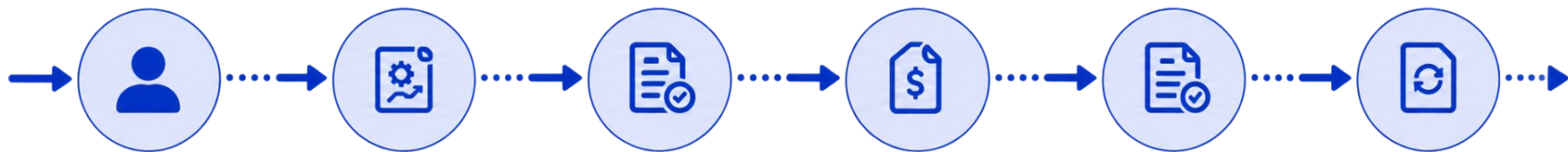
Data Model (Overview)



Data Quality Limitation: Customer-loan linkage is extremely limited, with only **0.11%** of **2013** loans successfully **matched** to customer records.

Methodology

A structured approach to assess portfolio health, risk, and borrower characteristics.



Data Understanding

Understand dataset structure, key attributes, relationships, and data quality.

Data Cleaning

Standardize customer IDs, handle missing values, and validate table relationships.

Portfolio Analysis

Assess portfolio OS, loan status composition, and TKB30 across cohorts.

Cohort & Risk Analysis

Identify weaker cohorts and concentrations of delinquent exposure.

Loan Characteristic Analysis

Evaluate grade, interest rate, loan amount, and their interaction with TKB30.

Dashboard & Recommendations

Translate findings into risk monitoring and lending recommendations.

Insight 1

2013 is a relatively weaker-performing cohort.

The **2013** cohort recorded the **lowest TKB30** at **96.49%**, while its **\$641.2K OS >30 days** represents meaningful delinquent exposure.



TKB30 (2013)

96.49%

Lowest among all cohorts.



OS > 30 Days (2013)

\$641.2K

Substantial delinquent exposure.

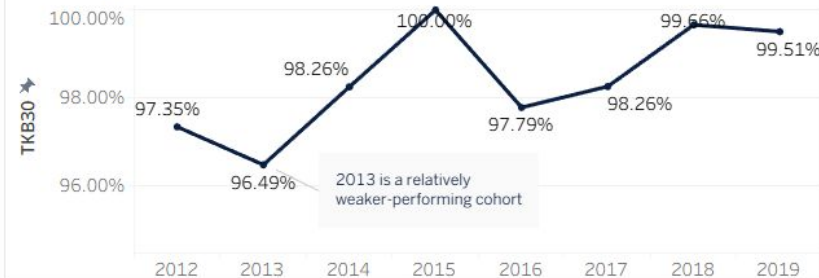


Total Loans (2013)

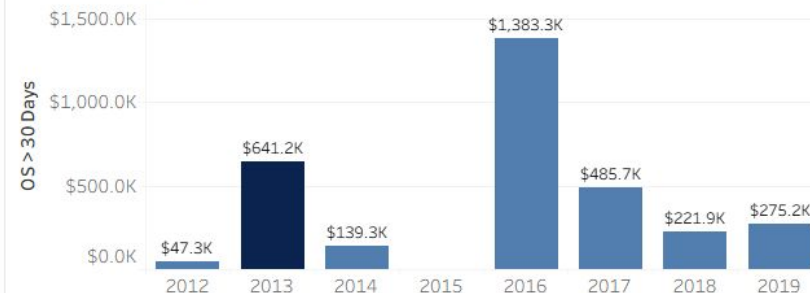
892

18.9% of total loans.

TKB30 by Cohort



OS > 30 Days by Cohort



Key Takeaway: The **2013** cohort underperforms others on **TKB30**, while its **\$641.2K OS >30 days** signals a meaningful risk exposure that warrants deeper analysis.

Insight 2

Risk exposure is concentrated in higher-risk loan segments.

Delinquent exposure is concentrated in the 20-25% interest-rate band and Grade E within the 2013 cohort.

**Risk Segment
OS > 30 Days**

\$308.1K

Interest Rate 20-25%

**Contribution to Total
OS > 30 Days (2013)**

48.06%

*From Grade D-G loans with
20-25% interest rate*

**Largest Grade
Contributor**

Grade E

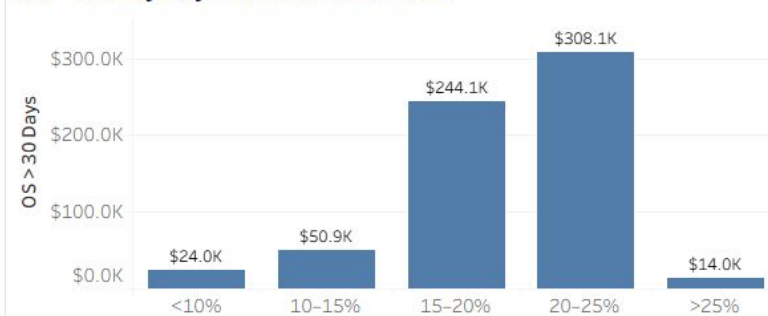
*Within the 20-25%
segment.*

**Grade E Contribution
(2013)**

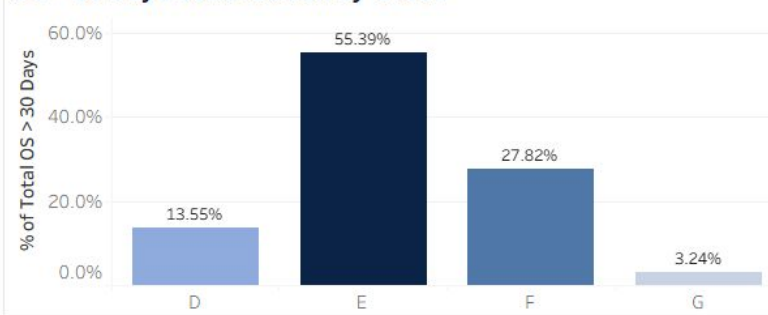
61.83%

*Of OS >30 days within this
segment.*

OS > 30 Days by Interest Rate Band



OS > 30 Days Contribution by Grade



Key Takeaway: Nearly half of the delinquent exposure comes from Grade **D-G** loans with **20-25%** interest rates, with Grade **E** contributing the **largest** share.

Insight 3

Larger loans show weaker performance within lower-grade segments.

Within Grade **E** and **F**, **TKB30** declines as loan amount increases, with the weakest performance observed in the **\$30K-\$40K** band.

Key Highlight
Grade **E** /
\$30K-\$40K loans

92.19%

TKB30
with **\$90K** OS >30 Days

TKB30 by Grade & Loan Amount Band

Grade	<\$10K	\$10K-\$20K	\$20K-\$30K	\$30K-\$40K
D	100.00%	86.43%	100.00%	100.00%
E		95.77%	93.76%	92.19%
F	100.00%	97.52%	94.72%	92.61%
G	100.00%	100.00%	100.00%	77.18%

Grade E: 95.77% → 93.76% → **92.19%** as loan amount increases.

Grade F: 97.52% → 94.72% → **92.61%**

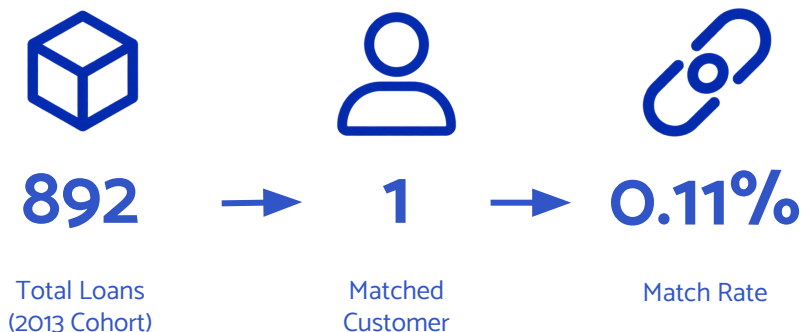


Key Takeaway: Within lower-grade loans, larger loan amounts are associated with weaker repayment performance, particularly for Grade **E** and **F**.

Insight 4

Customer-level risk profiling is constrained by data linkage.

Only a very small portion of loans can be linked to customer data, limiting the reliability of customer-level risk analysis.



Data Limitation

Customer demographics cannot reliably explain portfolio risk at the current linkage level.



Analytical Response

Prioritize loan-level risk characteristics for portfolio risk assessment.



Key Takeaway: Customer-level risk profiling is currently limited, making loan-level characteristics the more reliable basis for portfolio risk assessment.

Loan Portfolio & Credit Risk Dashboard

An interactive view of portfolio health, cohort performance, and credit risk exposure.



Portfolio Snapshot

Overview of key portfolio metrics and risk indicators.



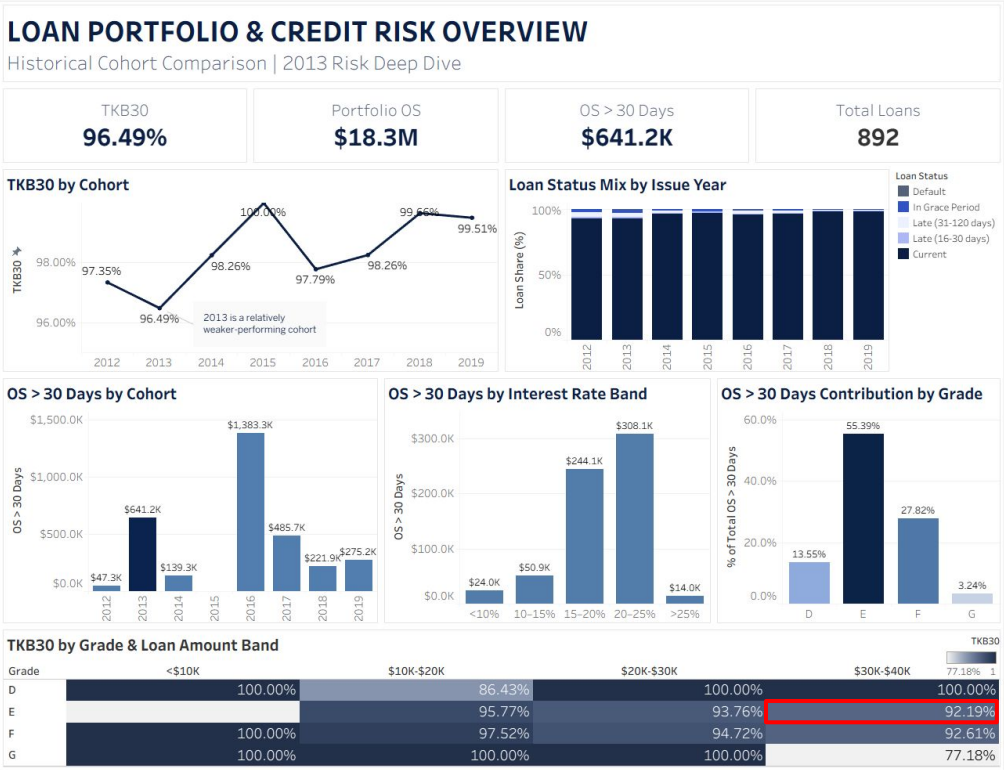
Cohort Monitoring

Compare portfolio performance across issue years.



Risk Segmentation

Analyze risk by interest rate, grade, and loan size.



Recommendations & Next Steps

Translating insights into actionable lending strategies.

01

Strengthen Risk-Based Underwriting

Apply stricter underwriting thresholds for lower-grade borrowers, especially for larger loan amounts and higher interest-rate segments.

02

Prioritize Exposure-Based Monitoring

Prioritize monitoring of segments with high OS >30 days contribution, especially Grade D–G loans within the 20–25% interest-rate range.

03

Introduce Early Delinquency Intervention

Establish early-warning monitoring for loans moving into delinquency to prevent further deterioration of portfolio quality.

04

Improve Customer–Loan Data Integration

Strengthen customer ID mapping and data integration to enable reliable borrower-level risk profiling in the future.



Next Steps: Validate risk thresholds with Risk Management, incorporate exposure-based monitoring into portfolio reviews, and improve customer–loan data integration for future borrower-level analysis.

THANK YOU

Explore the Project



TABLEAU DASHBOARD → [View Dashboard](#)



SQL ANALYSIS → [View SQL Queries](#)



evaarlina.github.io



[Eva Arliana](#)



evaarlina1@gmail.com